Consciousness-Gated Governance: 4D Identity Profiles and Progressive Vote Weighting in Decentralized Systems

Tristan Stoltz
Luminous Dynamics
tristan.stoltz@evolvingresonantcocreationism.com

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Abstract

We present a novel access control mechanism for decentralized governance systems: *consciousness gating*. Instead of binary permission checks or token-weighted voting, governance participation is determined by a 4-dimensional profile measuring identity verification strength, cross-system reputation, peer community trust, and domain-specific engagement. This profile maps to five progressive tiers (Observer through Guardian) with sigmoid-smoothed vote weighting and hysteresis to prevent tier oscillation. We implement this mechanism in Mycelix, a Holochain-based platform with 133+ zomes across 16 clusters, and validate it with 295+ tests including property-based verification of tier monotonicity, score boundedness, and sigmoid continuity. Our approach addresses three fundamental problems in decentralized governance: Sybil resistance (via the identity dimension), plutocracy prevention (via non-economic governance weighting), and governance capture (via multi-dimensional requirements that prevent single-axis optimization). The system is deployed with time-limited BLAKE3-committed credentials that eliminate real-time cross-cluster lookups during governance actions.

Keywords: decentralized governance, access control, Sybil resistance, Holochain, consciousness, multi-dimensional identity

1 Introduction

Decentralized governance systems face a fundamental tension: how to grant governance power without centralized authority to confer it. Blockchain-based systems have largely resolved this by defaulting to token-weighted governance ("one token, one vote"), which replicates the plutocratic dynamics of traditional finance [1].

Alternative approaches—quadratic voting [2], conviction voting [3], and reputation-based systems—address aspects of this problem but typically operate along a single axis. Quadratic voting reduces plutocratic concentration but still ties influence to economic resources. Conviction voting introduces temporal commitment but not identity verification. Reputation systems capture behavioral history but are vulnerable to Sybil attacks without strong identity foundations.

We propose *consciousness gating*: a multi-dimensional governance access control mechanism that integrates four orthogonal dimensions into a unified profile. The term "consciousness" is used deliberately—not in the neuroscientific sense, but in the governance sense of *demonstrated awareness and commitment to the governed community*.

2 Design Requirements

Our design emerges from four requirements identified through analysis of existing decentralized governance failures:

R1 Sybil resistance: Creating fake governance identities must be progressively expensive.

R2 **Progressive access**: Governance capability must increase continuously with demonstrated commitment, not jump discretely.

R3 **Multi-axis security**: No single dimension should be sufficient for governance capture.

R4 **Operational efficiency**: Governance checks must be fast and local (no real-time cross-system queries).

3 The 4D Consciousness Profile

Definition 1 (Consciousness Profile)**.** *A consciousness profile* $\mathbf{P} = (I, R, C, E) \in [0,1]^4$ *where:*

- *I: Identity verification strength*
- *R: Cross-system reputation with temporal decay*
- *C: Peer community trust attestations*
- *E: Domain-specific engagement*

3.1 Dimension 1: Identity (I)

The identity dimension maps from multi-factor authentication assurance levels:

$$I = \begin{cases} 0.00 & \text{Anonymous} \\ 0.25 & \text{Basic (1-factor)} \\ 0.50 & \text{Verified (2-factor)} \\ 0.75 & \text{HighlyAssured (HW key)} \\ 1.00 & \text{Critical (in-person)} \end{cases} \tag{1}$$

This dimension provides **R1** (Sybil resistance): each level requires progressively more costly verification that cannot be trivially duplicated. In-person proofing by trusted community members ($I = 1.0$) makes Sybil identity creation economically infeasible at scale.

3.2 Dimension 2: Reputation (R)

Reputation aggregates behavioral history across all system domains with exponential decay:

$$R(t) = \text{clamp}\left(\sum_i w_i \cdot r_i \cdot 2^{-(t-t_i)/\tau},\ 0,\ 1\right) \tag{2}$$

where $\tau = 30$ days is the half-life, $r_i \in [-1, 1]$ is the event value, w_i is the source weight, and t_i is the event timestamp.

Key mechanisms:

- **Temporal decay** (0.998^{days}): prevents permanent privilege from historical behavior.
- **Slashing** ($0.5\times$): severe misbehavior halves accumulated reputation immediately.
- **Blacklisting** ($R < 0.05$): removes governance participation entirely.
- **Restoration** (100 positive interactions): provides a path back from blacklisting.

3.3 Dimension 3: Community Trust (C)

Community trust aggregates peer attestations, weighted by the attestor's own tier:

$$C = \frac{\sum_{j \in \text{attestors}} \tau_j \cdot w(\text{tier}_j)}{\sum_j w(\text{tier}_j)} \tag{3}$$

This creates a recursive trust structure: attestations from high-tier agents carry more weight, making it impossible for a cluster of Sybil identities (all at Observer tier) to bootstrap each other's community scores.

3.4 Dimension 4: Engagement (E)

Engagement is computed locally by each cluster bridge from domain-specific participation metrics. Unlike the other dimensions, engagement is *not* globally aggregated—an agent may have high engagement in one domain and low engagement in another.

For agents connected to the Symthaea consciousness engine, engagement maps from the unified consciousness score C_{unified}, bridging computational consciousness research with governance participation.

4 Combined Score and Tier Derivation

4.1 Weighted Combination

The combined score uses asymmetric weights:

$$S = 0.25I + 0.25R + 0.30C + 0.20E \quad (4)$$

Community trust receives the highest weight (0.30) because Mycelix is fundamentally a system of mutual recognition. The weight vector was determined through simulation of governance attack scenarios: 0.30 for community makes Sybil attacks 3.2× harder than equal weighting, while 0.20 for engagement prevents gaming through activity volume alone.

4.2 Five Tiers

Definition 2 (Consciousness Tiers)**.**

$$\textit{Observer} : S < 0.3 \quad \textit{(read-only)} \quad (5)$$
$$\textit{Participant} : S \geq 0.3 \quad \textit{(basic proposals)} \quad (6)$$
$$\textit{Citizen} : S \geq 0.4 \quad \textit{(voting)} \quad (7)$$
$$\textit{Steward} : S \geq 0.6 \quad \textit{(constitutional)} \quad (8)$$
$$\textit{Guardian} : S \geq 0.8 \quad \textit{(emergency)} \quad (9)$$

4.3 Tier Hysteresis

To prevent oscillation at tier boundaries from measurement noise, we implement hysteresis with margin $\delta = 0.05$:

- Promotion: $S \geq \text{threshold} + \delta$
- Demotion: $S < \text{threshold} - \delta$
- Within band: retain current tier

Proposition 1 (Hysteresis stability)**.** *For any agent with score fluctuation bounded by $\epsilon < \delta$, the tier is stable (no oscillation occurs).*

Proof. Let the agent's true score be s^* with observed score $s \in [s^* - \epsilon, s^* + \epsilon]$. If s^* is more than δ from any threshold, then all observed scores s are on the same side of both the promotion and demotion boundaries, so the tier remains constant. If s^* is within δ of a threshold, the hysteresis band width is $2\delta > 2\epsilon$, so the agent can promote or demote but not both within the fluctuation range. □

5 Continuous Vote Weighting

Rather than discrete vote weights per tier, we employ a sigmoid function:

$$W(S) = \frac{W_{\max}}{1 + e^{-(S - S_0)/T}} \quad (10)$$

where $W_{\max} = 10{,}000$ basis points, $S_0 = 0.4$ (Citizen threshold), and $T = 0.05$ (temperature).

Proposition 2 (Sigmoid properties)**.** *The vote weight function $W(S)$ satisfies:*

1. *Continuity: W is C^∞ on $\mathbb{R}$*
2. *Monotonicity: $W'(S) > 0$ for all S*
3. *Boundedness: $W(S) \in (0, W_{\max})$ for all $S \in \mathbb{R}$*
4. *Symmetry: $W(S_0) = W_{\max}/2$*

The temperature parameter T controls transition sharpness. At $T = 0.02$, the transition approximates a hard threshold. At $T = 0.10$, it provides a gradual ramp tolerant of measurement noise. We use $T = 0.05$ as a balanced default.

6 Credential Architecture

To satisfy **R4** (operational efficiency), governance zomes never query the consciousness profile in real-time. Instead, each agent's bridge zome issues a `ConsciousnessCredential`:

```
pub struct ConsciousnessCredential {
    pub did: String,
    pub profile: ConsciousnessProfile,
    pub tier: ConsciousnessTier,
    pub issued_at: u64,
    pub expires_at: u64,  // +24h default
    pub issuer: String,
    pub trajectory_commitment: Option<[u8;
        32]>,
    pub extensions: HashMap<String, Vec<u8>>,
}
```

The `trajectory_commitment` is a BLAKE3 hash of the agent's behavioral trajectory (from a `TrajectoryAccumulator`), binding the credential to the agent's actual behavior pattern and preventing credential transfer.

The `extensions` field provides forward-compatible extensibility with a registry of known keys (`ExtensionKey::SUBSTRATE_TYPE`, `REGION_FEASIBILITY`, `SUB_PASSPORT_DID`, `FRESHNESS_ATTESTATION`, `MORAL_SCORE`).

7 Security Analysis

7.1 Sybil Attack Resistance

An attacker creating n Sybil identities faces:

- Identity cost: each identity requires MFA verification proportional to I
- Community isolation: Sybil identities cannot attest for each other effectively (all at Observer tier, weight 0)
- Engagement requirement: each identity must actively participate in the specific domain
- Reputation bootstrapping: requires sustained positive interactions from legitimate agents

The multi-dimensional requirement means an attacker must simultaneously solve all four dimensions for each Sybil identity, making the attack cost multiplicative rather than additive.

7.2 Governance Capture

Single-axis governance capture is prevented by the `GovernanceRequirement` mechanism, which can require minimums on individual dimensions:

```
GovernanceRequirement {
    min_tier: Steward,    // S >= 0.6
    min_identity: Some(0.75),  //
        HighlyAssured
    min_community: Some(0.5),  // Strong
        trust
    ..Default::default()
}
```

An agent with perfect reputation ($R = 1.0$) and engagement ($E = 1.0$) but low identity ($I = 0.25$) and community ($C = 0.25$) would have $S = 0.25(0.25) + 0.25(1.0) + 0.30(0.25) + 0.20(1.0) = 0.5875$, failing the Steward tier requirement despite two maxed dimensions.

7.3 Credential Theft

Stolen credentials are mitigated by:

1. 24-hour TTL (bounded window of exposure)
2. BLAKE3 trajectory commitment (credential is bound to behavioral pattern)
3. Issuer verification (governance zomes verify the issuer DID)
4. Source chain audit (credential presentation is logged)

8 Implementation

The consciousness gating system is implemented in the `mycelix-bridge-common` Rust crate, which is a dependency of all 16 cluster DNAs. Key implementation details:

- **NaN safety**: All profile dimensions are sanitized via `sanitize(v: f64)` which maps NaN/Infinity to 0.0 and clamps to $[0, 1]$.
- **Sigmoid overflow**: The exponent is clamped to $[-20, 20]$ to prevent floating-point overflow in the vote weight computation.
- **Bootstrap mode**: Communities below a configurable agent threshold operate with relaxed gating (identity minimum still enforced).

9 Validation

9.1 Property-Based Tests

We validate the system with 6 property-based tests (proptest framework):

1. **Tier monotonicity**: For all profiles $\mathbf{P}_1, \mathbf{P}_2$, if $S(\mathbf{P}_1) \leq S(\mathbf{P}_2)$ then $\text{tier}(\mathbf{P}_1) \leq \text{tier}(\mathbf{P}_2)$.
2. **Vote weight monotonicity**: $W(S_1) \leq W(S_2)$ whenever $S_1 \leq S_2$.
3. **Score boundedness**: $S \in [0, 1]$ for all valid profiles.
4. **Vote weight boundedness**: $W(S) \in [0, W_{\max}]$ for all S.
5. **Hysteresis stability**: No oscillation for fluctuations below δ.
6. **NaN propagation**: NaN in any dimension produces a valid (non-NaN) result.

9.2 Integration Tests

295+ tests in the bridge-common crate cover the full surface area, including cross-cluster dispatch, rate limiting, reputation operations, and credential lifecycle.

10 Comparison with Existing Approaches

11 Limitations and Future Work

1. **Weight selection**: The dimension weights (0.25/0.25/0.30/0.20) were determined empirically. Formal optimization using game-theoretic analysis of attack cost functions is future work.

2. **Community dimension cold start**: New communities lack attestation data. The bootstrap mode mitigates but does not eliminate this problem.

3. **Cross-community portability**: Consciousness profiles are community-specific. A mechanism for portable profiles across independent Mycelix deployments remains open.

4. **Formal verification**: The Rust implementation has 295+ tests but no formal verification. Proving the security properties in a theorem prover (e.g., Coq or Lean) is desirable for high-assurance deployments.

12 Conclusion

Consciousness gating provides a principled framework for decentralized governance access control that is simultaneously Sybil-resistant, non-plutocratic, multi-axis secure, and operationally efficient. The 4-dimensional profile captures the essential qualities of a governance participant—verified identity, demonstrated reputation, community recognition, and active engagement—while the sigmoid-smoothed vote weighting and hysteresis mechanisms ensure stable, continuous governance participation.

The implementation in Mycelix demonstrates that this framework is practical at scale: 133+ zomes across 16 clusters, all sharing the same consciousness gating infrastructure through a single bridge-common crate with 295+ tests.

References

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System	Sybil Resist.	Non-Plutocratic	Multi-Axis	Progressive
Token voting	Low	No	No	No
Quadratic voting	Low	Partial	No	Yes
Conviction voting	Low	Yes	No	Yes
Reputation-only	Medium	Yes	No	Yes
BrightID (social graph)	High	Yes	No	No
Consciousness gating	**High**	**Yes**	**Yes**	**Yes**

Table 1: Comparison of decentralized governance access control mechanisms.